

Down Syndrome

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INTRODUCTION

Down syndrome (DS) or trisomy 21 is one of the most common causes of intellectual disability (ID) and recent prevalence estimates suggest that there are 11.8¹ to 14.47² per 10,000 live births in the United States with DS. It is estimated that 250,000 people in the United States have DS. In turn, World Health Organization estimates for DS range between 10 to 11 in 10,000 live births worldwide.³ This chapter summarizes current knowledge on DS risk and prevalence, behavioral, psychiatric, physical and neurological manifestations of the disease, and the effects of aging on cognitive functioning of individuals with DS.

HALLMARKS OF DOWN SYNDROME

There are an array of features that may be present in DS. Ten hallmark symptoms seen in newborns have been identified to assist in postnatal confirmation of DS,⁴⁻⁶ and include: flat facial profile, slanted palpebral fissures, anomalous ears, hypotonia, poor Moro reflex, dysplasia of midphalanx of the fifth finger, transverse palmar crease, excessive skin at the nape of the neck, hyperflexibility of joints, and dysplasia of pelvis. Other common features that may also present are: small head, short neck, protruding tongue, unusually shaped eyes, Brushfield spots in the eye, relatively short fingers, space between first and second toes (sandal gap), narrow palate, slow growth, shorter stature, late developmental milestones, and ID (typically mild to moderate, with IQ in the 50–70 or 35–50 range, respectively). Symptoms associated with DS may include: heart defect (50% of children), digestive anomalies, hypothyroidism, celiac disease, gastroesophageal reflux, childhood leukemia, and behavioral issues such as attention problems, obsessive/compulsive behavior, stubbornness, tantrums, autism, and dementia (after age 50).

INHERITANCE

Maternal age is a primary risk factor for DS. The chances of a woman having a child with DS increase with age because older eggs have a higher risk of improper chromosome division. By age 35, a woman's risk of conceiving a child with DS is 1 in 400; by age 45 the risk is 1 in 35. However, most children with DS are born to women under the age of 35, as they are responsible for the majority of all births. Having had one child with DS leads to a marginal increase of 1% risk of having another child with DS.⁷

Health of the mother prior to and during pregnancy has also been linked to DS risk. Mothers of DS children tended to have more significant illnesses before conception and more medication ingestion the year before conception.⁸ Women who suffered from gestational diabetes were twice as likely to have offspring with chromosomal abnormalities, including DS, than women who did not.⁷

Timing of pregnancy is thought to have some bearing on DS. Short intervals between pregnancies has been linked to increased DS occurrence.⁹ Jongbloet proposed that short periods of anovulatory activity followed by conception

correlate with greater occurrence of DS.^{10,11} It is suggested that conceptions during the transition between anovulation and the re-establishment of regular ovulation may be a time vulnerable to maternal meiotic nondisjunction.

Last, some genetic issues may increase risk of DS. One such example is that a parent being a carrier of the genetic translocation for DS increases risk of DS. It has also been shown that families with histories of Alzheimer disease (AD) are more likely to have offspring with DS.¹²

DIAGNOSIS AND TESTING

First-trimester screening involves offering prenatal testing if the maternal age is 35 years or older or if maternal serum markers in combination with ultrasound measures of fetal nuchal translucency are suggestive.¹³ Subsequently, the primary method for identifying DS genetically is through a karyotype, which requires 10–14 days for test results as cell cultures are required. Additionally, using FISH (fluorescence *in situ* hybridization using a probe for chromosome 21) or polymerase chain reaction (PCR) with amniotic fluid samples, lymphocytes from blood or buccal mucosa cellular preparations is a more rapid approach and appropriate for establishing mosaicism. Karyotyping and FISH/PCR each have unique advantages and disadvantages, as discussed by Gekas and colleagues,¹⁴ but a consensus is growing that using both FISH and karyotyping provides the highest accuracy.¹⁵

Gene Identification

DS was initially described by J. Langdon Down in 1866¹⁶ and identified as a chromosome 21 trisomy by Lejeune in 1959.¹⁷ Called nondisjunction trisomy 21, this type of DS is caused by abnormal cell division and results in three copies of chromosome 21. Nondisjunction accounts for 95% of all DS cases. Two additional types of DS have also been identified. Partial trisomy 21 is a rare form of DS (estimates range from 1–5% from a few case reports)^{18–20} and involves the triplication of a piece of chromosome 21 when part of chromosome 21 breaks off during cell division and becomes attached (translocated) to another chromosome (usually chromosome 14). The total number of chromosomes in the cells is still 46, however, the additional part of chromosome 21 causes the characteristics of DS. The second additional cause of DS is mosaicism (which affects approximately between 1–4% of individuals), where an extra chromosome 21 is present in some but not all cells in the body and typically the phenotype is milder than full trisomy 21.^{21–23} Interestingly, the severity of the DS phenotype is linked to the percentage of mosaicism.²³ Heredity is not a factor in trisomy 21 (nondisjunction) or mosaicism. Yet, in one-third of cases of DS resulting in a translocation, there is a hereditary component.

Disease Identification

Chapman and Hesketh²⁴ have written a comprehensive review of the behavioral phenotype in DS for children and adolescents that is characterized by ID, specific deficits in expressive language development, impaired speech intelligibility, and impaired verbal short-term memory (STM). Adaptive behavior is consistent with general intelligence. Levels of maladaptive behavior are lower than for comparison groups with ID and do not change significantly with age. In DS adults, the conditions that most affect behavior are depression, hypothyroidism, and dementia.²⁴

EARLY INTERVENTION/TREATMENT

At the current time, children with DS are almost always referred for early intervention (EI) programs after birth.²⁵ The most common EI services for babies with DS are physical therapy (PT) and speech therapy (ST). PT for children with DS focuses on motor development and improving muscle tone. ST is important for children with DS because they frequently have small mouths with enlarged tongues, and, therefore, have trouble speaking clearly. Further, hypotonia also inhibits the movement of the facial muscles required for speech, and hearing problems inhibit the ability to hear and mimic speech. ST focuses on learning to communicate clearly through talking and frequently through sign language.

The effectiveness of EI has been investigated. Connolly and colleagues²⁶ compared the intellectual and adaptive functioning of children with DS who participated in an EI program to those with DS who did not participate in an EI program. The mean IQ for the EI group was found to be 12 points higher than for the comparison group ($p < .005$, a significant difference). The social quotient (SQ), measured by the Vineland Social Maturity Scale, for the EI group

was also found to be significantly higher by 9 points ($p < .05$) than the comparison group. Children in the EI group met developmental milestones earlier than the comparison group, though not as early as children without ID. The authors suggest that EI during early infancy bolsters the development of intellectual and adaptive skills during early childhood. Other research shows that the effects of EI have lasting benefit years after the end of treatment.²⁷

EI through medical treatment has likely been a major factor that has helped improve health and quality of life of children with DS. Corrective surgery for heart defects, gastrointestinal irregularities, screening for visual impairment, ear infections, hearing loss, hypothyroidism, and obesity are amongst the early medical treatments individuals with DS can benefit from given the comorbidities associated with their condition.²⁸

PREVALENCE

The prevalence of DS births has been increasing. A cross-sectional analysis of live-born infants with DS during 1979–2003 from 10 regions of the United States found the prevalence of DS at birth increased by 31%, from 9.0 to 11.8 per 10,000 live births. DS births over time were found to be increasing in proportion to the increasing rate of mothers aged 35 or older giving birth in the studied regions. These findings are in line with findings from prior research.¹ In the United States, rates of DS vary by race/ethnicity. The prevalence of DS at birth is significantly lower among non-Hispanic blacks and was higher among Hispanics compared to the prevalence rates among non-Hispanic whites. Hispanic infants were found to have higher rates of DS than other infants, even when differences in maternal age were accounted for.²⁹ Rates for Hispanic, white and African American infants were, 11.8, 9.2 and 7.3, respectively, per 10,000 live births.²⁹ Differences seen may be due to contrasting use of prenatal screening. Use of prenatal diagnosis and termination of pregnancy significantly reduced the prevalence of DS births among white women, but not among women of other races.³⁰ However, this finding was not supported in another study.³¹

DISEASE EVOLUTION

Along with longer life expectancy comes a larger population of adults with DS who display premature age-related changes in their health.³² A key challenge for adults with DS as they age is the increasing risk for developing clinical symptoms of Alzheimer disease (AD). (For a review of risk factors for dementia in DS see ^{33,34}.)

Virtually all adults with DS (full trisomy 21) over the age of 40 years have sufficient amyloid- β plaques and neurofibrillary tangles for a neuropathological diagnosis of AD.^{35–37} This is due to the overexpression of the amyloid precursor protein gene on chromosome 21, which is cleaved to produce amyloid- β . The early signs of dementia in people with DS might be a result of dysfunction of the frontal lobe and hippocampus—areas in which amyloid- β first accumulates during the early stages of dementia.³⁸ However, despite the presence of AD neuropathology, not all adults with DS show clinical signs of dementia. The prevalence of clinical dementia increases with age over 50 years, reaching over 75% in people with DS aged over 65 years.^{33,38}

Noetzel³⁹ suggests that the phases of clinical deterioration in people with DS and dementia can involve memory impairment in the initial stages with reduced verbal output in those with mild–moderate ID. In more severe ID, the first symptoms of dementia may be more behavioral in nature, including apathy, inattention, decreased social interaction, and spatial disorientation. In the middle-severity phase, slowed gait, shuffling, loss of self-help skills, and increased seizure activity can occur.⁴⁰ In the final phase of dementia, individuals become nonambulatory and bedridden, with onset of spasticity and pathological release reflexes.^{40–44}

Diagnosis of Dementia in DS

Deb and Braganza⁴⁵ compared diagnosing dementia in individuals with DS using the International Classification of Diseases, 10th edition (ICD-10) clinician rating, Dementia Scale for Down Syndrome (DSDS), Dementia Questionnaire for Persons with Mental Retardation (DMR), and Mini-Mental State Exam (MMSE). It was found that clinician rating, DSDS and DMR had high levels of concordance in terms of diagnosing dementia. Advanced AD was readily diagnosed in the study. Difficulties with diagnosis arose when individuals with DS were suspected to be in early stages of dementia. In such cases, rating scales and clinician report were discrepant. MMSE did not effectively diagnose dementia in DS in the majority of cases. Strydom et al.⁴⁶ found that dementia was common in older individuals with ID, but prevalence differed according to the diagnostic criteria being used. These authors found that the Diagnostic and Statistical Manual of Mental Disorders, 4th edition, (DSM-IV) criteria were more inclusive, while ICD-10 excluded individuals with even moderate dementia.

Current evidence suggests that behavioral and personality changes may be the first and most reliable indicators of cognitive decline in DS. Research by Holland and colleagues⁴⁷ demonstrated that when informants first reported symptoms of change in individuals with DS they were predominantly behavioral or psychiatric, consistent with frontal lobe dysfunction. This suggests that functions served by the frontal lobes may be the first compromised with the development of AD-like neuropathology in people with DS.⁴⁸ In a 14-year follow-up study of 77 women with DS over the age of 35 years, it was found that amongst the measures used to examine cognitive functioning, the informant-based Dementia Questionnaire for Mentally Retarded Persons (DMR) was found to show decline approximately 5 years prior to diagnosis of dementia.⁴⁹ Therefore, honing in on the behavioral observations of caregivers of older individuals with DS to detect change may prove to be important in early identification of cognitive decline. Standardized evaluation of behavioral/psychiatric issues could contribute to the ability to diagnose dementia early in adults with DS. Better understanding of the pattern of progression of dementia in individuals with DS may help to improve diagnostic accuracy and allow for earlier treatment for those affected.⁵⁰

As psychiatric symptoms are prevalent features of dementia in the population with DS and may appear prior to substantial changes in daily functioning, the specific types of psychiatric symptoms common in AD in those with and without DS have been evaluated to examine potential differences in presentation of symptoms in these populations.⁵⁰

The most common types of neuropsychiatric symptoms in AD in the general population are a) delusions, b) hallucinations, c) agitation, d) aggression, e) depression, f) anxiety, and g) apathy. Study of psychiatric and behavioral symptoms in adults with DS has been conducted^{51,52,53} and found that generally, adults with DS without dementia had a low prevalence of delusions, hallucinations, behavioral problems, or depression coupled with high levels of apathy. Individuals with DS in the questionable cognitive status group were much more likely to have delusions and exhibit verbal and physical violence than those with no dementia. The authors suggest these aforementioned behaviors may be early indicators of dementia onset.⁵⁰ For individuals with DS diagnosed with dementia, delusions, visual hallucinations, misidentifications, and depression were prevalent, persistent and less amenable to correction.

Recently, Lott and colleagues⁴⁰ investigated the occurrence of seizures as another early indicator of dementia onset in individuals with DS. Prior research indicates that individuals with DS over the age of 45 years with new-onset seizures are more likely to develop AD and seizures may demarcate the beginning of dementia.^{54,55} In a sample of 53 adults with DS and AD, it was found that those with seizures were more likely to become untestable on cognitive measures (used as a surrogate for severe impairment) than those without seizures, showing a more marked cognitive decline. The authors suggest that high brain levels of amyloid- β peptides in individuals with DS and AD are likely interfering with neuronal and synaptic activity, thereby lowering the seizure threshold. Taken together, the presence of seizures in individuals with DS may be an early sign that cognitive decline is impending, and may be useful in diagnosis of dementia in DS.

Nieuwenhuis-Mark⁵⁶ reviewed the issues involved with diagnosing AD in DS and provided recommendations which include: annual screening for dementia for all people with DS age 35 and over; modified classification systems to capture early and atypical symptoms of dementia; and developing repeatable standardized assessment batteries to be utilized for diagnosing dementia in DS. Further, assessments of individuals with DS should also include a clinical history and physical examination⁵⁷ and be sensitive to psychiatric, behavioral, and mood changes, while taking into account the sensory and physical deficits that are more likely to occur at older ages.⁵⁸

In suggesting a test battery for diagnosing dementia in individuals with ID, Burt and Aylward⁵⁹ posit that dementia should be diagnosed only when longitudinal data demonstrate clinically significant declines in functioning. To meet diagnostic criteria for dementia, documented declines on at least one memory test and one other test of cognitive ability are needed. Changes on cognitive tests should also coincide with changes in everyday functioning.

Although a specific benchmark test or test battery that can reliably diagnose dementia in DS across ID levels and situations has yet to be endorsed as the standard, Edgin and colleagues⁵⁸ have developed the Arizona Cognitive Test Battery (ACTB) to assess the cognitive phenotype in DS. ACTB utilizes tasks from the Cambridge Neuropsychological Testing Automated Battery (CANTAB)⁶⁰ to test general cognitive abilities, prefrontal, hippocampal and cerebellar functions. The developers of the battery have found that the ACTB provided consistent results across contexts, socioeconomic backgrounds and ethnicities, and was positively correlated with informant reports. With further investigation, the ACTB may become a beneficial tool for following changes in individuals with DS by helping to identify the specific brain regions affected as cognition changes over time.

End of Life: Mechanisms and Comorbidities

Common causes of death in the DS population include leukemia (in childhood), respiratory illness, congenital circulatory defects, diseases of the digestive system, dementia and AD.³² The likelihood of death from a particular

comorbidity varies by age. Looking at comorbidity and mortality in DS across the lifespan it has been determined that across all age categories pneumonia and other respiratory infections were the most common cause of death ranging from 23% in adulthood to 40% in senescence.⁶¹ With attention to specific stages:

- Childhood to early adulthood (ages 0–18 years): coronary heart disease (CHD)=13% of deaths
- Adulthood (ages 19–40 years): CHD=23% of deaths
- Senescence (ages 40+): coronary artery disease=10% of deaths; cardiac, renal, and respiratory failure=9% of deaths

Race has been shown to be a factor in mortality in individuals with DS in the United States.^{62,63} Review of death certificates found that the median age at death increased from 25 years in 1983 to 49 years in 1997 for white individuals with DS. However, there were apparent racial disparities in median age at death. For blacks the median age at death was 25 years, and for people of other races only 11 years. This may be due to differences in the manifestations of symptoms in DS in individuals of other races, as it was found that CHD and pulmonary circulation were reported more frequently on the death certificates of people of other races with DS than on those of whites with DS.

With reference to DS adults, age, dementia status, and mobility restrictions were found to be the most important predictors of mortality in a study of 500 people with DS age 45 and older.⁶⁴ Studies investigating whether prior levels of functional abilities or declines in functional abilities predict mortality in adults with DS have shown mixed results. Some researchers have found that declines in functional abilities and cognition were related to increased mortality,^{64,65} while findings of other studies have not found this relationship.^{66–68} Table 15.1 provides a synopsis of the current evidence of the dementia health risks in DS dementia.

TABLE 15.1 Dementia Health Risks and Down Syndrome Dementia

Dementia health risk factor	Risk and putative mechanism	Down syndrome
Hypertension	Stroke; cerebrovascular disease; protein extravasation	Individuals with DS have lower resting heart rates and lower blood pressure than general population
Obesity	High BMI is associated with a 59% increased risk for AD; also a risk for sleep apnea syndrome (see below)	45–79% of males; 56–96% of females are reported to be overweight
Diabetes	May alter A β clearance in brain; may promote inflammation	Age of onset ~22 years for type 1 diabetes is comparable to general population; preliminary data on type 2 diabetes suggests a lower rate, however
Cardiovascular disease	Promotion of cerebrovascular disease; associated dyslipidemia increases risk of brain plaque pathology	Rate of mitral valve prolapse is high However, lower risk for cardiovascular disease in adults with DS compared to general population; this includes lower rates of hypercholesterolemia and heart disease compared to adults with other intellectual disabilities
Cerebrovascular disease	Direct injury to brain regions involved in cognition; inflammation; hypoperfusion; increased A β production	Lower risk for cerebrovascular disease observed in adults with DS compared to general population
Head injury	A β and tau pathologies are increased in brain; increased APP production	No available epidemiological reports
Sleep apnea	Lowered oxygen during sleep impacting brain	An estimated 94% of persons with DS, ages 17–56 have obstructive sleep apnea of varying severity
Thyroid dysfunction	May reflect thyroid stimulating hormone on A β processing; a cofactor for vascular dementia risk	Seen in 35–40% of adults with incidence increasing with advancing age; Hashimoto thyroiditis may be mistaken for dementia
Seizures	Seen with neurodegeneration, early onset, and ApoE ϵ 4 in ~2–8% of persons with AD	Possible link to myoclonus epilepsy gene on chromosome 21; rate increases with age in DS from 7–46% (over age 50) up to 84% of persons with DS and dementia

Abbreviations: A β , amyloid- β peptide; APP, amyloid precursor protein; BMI, body-mass index.
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PATHOPHYSIOLOGY

Postmortem observations and volumetric magnetic resonance imaging (MRI) studies show that people with DS have reduced brain volumes and brachycephaly with disproportionately smaller volumes in frontal and temporal areas and the cerebellum compared with healthy individuals. While in contrast, subcortical areas have relatively normal brain volumes. The parahippocampal gyrus appears larger on MRI in people with DS than in healthy individuals, and neuropsychological tests assessing function in parahippocampal and perirhinal regions suggest that these areas are functioning normally.³⁸ In relation to aging and brain structure, a recent investigation utilized fractional anisotropy (FA) from diffusion tensor imaging (DTI) to detect changes in white matter (WM) integrity between adults with DS without dementia, from adults with DS with dementia and from age-matched non-DS adults.⁶⁹ FA in conjunction with scores on cognitive testing (Brief Praxis Test and the Severe Impairment Battery) demonstrate significantly lower WM integrity in DS as compared to non-DS, particularly in the frontal lobes, corpus callosum, and association tracts predominantly within the frontoparietal regions. Comparing DS with and without dementia, DTI shows significant clusters where DS with dementia have lower FA than DS without dementia. The authors suggest that, taken together, developmental reductions in frontal cortex FA in DS that are present prior to dementia leave this area more vulnerable to the effects of brain aging and the progression to dementia.

Morphometric studies of the cortex in people with DS show fewer neurons, decreased neuronal densities, and abnormal neuronal distribution. The synaptic density, synaptic length, and synaptic contact zones are abnormal. These neuroanatomical abnormalities might be associated with the learning and memory deficits.³⁸

The cognitive profile in DS shows deficits primarily in morphosyntax, verbal short-term memory (STM), and explicit long-term memory (LTM), while visuospatial STM, associative learning, and implicit LTM are typically preserved.³⁸

Other research has shown a relative sparing of basic STM skills in children with DS, a strength that decays when greater demands such as increased memory load or executive requirements are placed onto the children.⁷⁰ This investigation also indicated that children with DS were extremely impaired on measures sensitive to hippocampal dysfunction,⁷¹ a structure critically impaired in the DS brain phenotype.⁷² Taken together, linking behavioral results to the neurobiological underpinnings of the visual-spatial memory in children with DS, the relative sparing of spatial STM performance could be sustained by the documented preservation of parietal lobe gray matter.⁷³ The (pre)frontal abnormalities frequently documented in the study of the DS brain phenotype⁷³ could underpin poor performance in the strategic self-ordered task. Tasks that are sensitive to medial temporal lobe, and in particular to hippocampal functioning (pattern and spatial recognition), show poor performance as a consequence of the disproportionate impairment of this structure in the DS brain.^{70,72,73}

In contrast, Pennington and colleagues⁷² tested prefrontal and hippocampal functions in a sample of school-aged individuals with DS compared with a sample of typically developing children individually matched on mental age (MA). The group with DS performed worse than MA controls on each hippocampal measure but not on any of the prefrontal measures. These authors suggest the pattern of results provides evidence for dissociation between two neuropsychological functions: hippocampal-mediated LTM and prefrontally mediated working memory. Thus, there is some work that remains to be done with regard to how cognition in DS is measured so that the field can make sense of the discrepant research findings. (For detailed review of neural and cognitive features of DS see ⁷⁴).

CONCLUSIONS

DS is one of the most common genetic diseases and its prevalence is increasing with the growing rates of women over the age of 35 years giving birth. DS brings with it a host of implications for the health of affected individuals at all stages of the lifespan. EI at younger ages has shown to benefit DS children by early identification and amelioration of congenital defects, improved sensory capabilities, and development of better communication skills. Improved health in early life results in increased longevity. Thus, there are more DS individuals living into older age, and experiencing age-related problems such as AD. Better understanding of the cognitive, psychiatric, and neuropathological brain changes that occur in older adults with DS will help increase early detection of cognitive decline and ultimately may lead to improved quality of life at the end of the lifespan.

ACKNOWLEDGEMENTS

Funding provided by NIH/DHHS Eunice Kennedy Shriver National Institute of Child Health & Human Development R01 HD064993.

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